EvoSafeHarness: Evolving Model- and Domain-Specific Harnesses for Securing Agents

Nanxi Li, Yingzi Ma, Yulong Cao, Chong Xiang, G. Edward Suh, Bo Li, Yejin Choi, Dawn Song, Chaowei Xiao · arXiv 2609.05903v2

HarnessRSI 68 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Searched object: harness H = (P, C) around a frozen model

- **P** NL policy bundle — target + guard-judge policies
- **C** Executable code — pre/post tool-call hooks + state

1 Designer proposes H
free-form (P, C) from the domain spec + archive of scores and traces

2 Criticizer, fresh context
sees H + spec, never search tasks; flags rules a rename would evade

3 4-stage cascade
static → smoke (3) → gate (12; stop if bound < −10 pts) → full (30)

4 Analyzer
failure-pattern and design-lesson logs, per channel, go back to the archive

loop

after a fixed budget: best U − ASR on search data
held-out tasks only for the final evaluation;
one harness per model × domain, model weights frozen

2 Headline: lower ASR, small utility cost

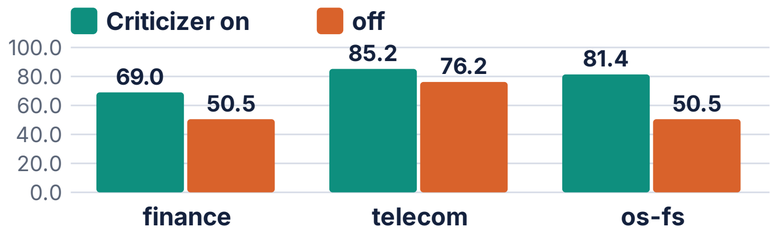
DecodingTrust-Agent, 15 searched cells (Table 1 average)

	Utility ↑	Direct ASR ↓	Indirect ASR ↓
No defense	83.1	50.9	40.4
CaMeL	69.7	44.3	31.1
DRIFT	77.1	49.3	35.2
Progent	56.4	14.7	6.3
Evo (ours)	79.8	12.6	7.4

One search per cell; held-out: 30 benign + 35 direct + 35 indirect tasks per domain. Baselines are frozen, not searched. Evo = EvoSafeHarness.

3 Controls that decide if the gain is real

Criticizer off vs on · held-out score (Kimi-K2.5)



Search-set score was equal or higher without it; −19.5 held-out on average. One search per arm; off-arm os-fs 50.5 vs primary-grid search 51.43.

Held-out score	with	without
Warm start · fin	69.05	59.52
Warm start · tel	85.24	73.33
Warm start · os-fs	81.43	71.90

- **AgentDojo** util / ASR: Evo 82.8 / 0.0, Progent 75.3 / 4.8, DRIFT 74.6 / 1.7, CaMeL 41.0 / 0.0, undefended 71.5 / 27.4. **Frozen onto AgentDyn:** Evo 75.0 / 0.0, undefended 73.3 / 12.8
- vs adapted Meta-Harness search (ASB, one search each): 61.9 vs 47.3; its budget is not stated
- Cascade: −43.6% tokens vs the staged schedule, same pick in 22 of 22 banks (offline replay)
- McNemar: ASR lower in 13 of 15 cells; utility CIs overlap in all 15 (30 benign tasks per cell)
- **Not reported:** search for baselines, matched compute, same-configuration repeat searches

4 Where the gain is absent or small

Util · D-ASR · I-ASR	Progent	Evo
os-fs	42.7 · 6.3 · 2.3	76.0 · 24.6 · 8.0
finance	56.0 · 4.6 · 6.9	84.6 · 9.7 · 13.7
telecom	70.7 · 33.1 · 9.7	78.6 · 3.4 · 0.6

- **Progent**: lower ASR on os-fs and finance
- DeepSeek-V4-Flash os-fs: direct ASR still **74.3** (undefended 94.3)
- Of 105 residual successes, 57 are os-fs; client scams 9/15 (60.0) unchanged
- ASB: Kimi-K2.6 utility 65.3 vs 66.7 undefended; GLM-5.2 ASR ties SafeHarness (8.6)

Takeaway

A per-deployment search with a fresh-context Criticizer cuts DTAP attack success far below CaMeL and DRIFT at a 3.3-point utility cost. Progent has lower ASR on os-fs and finance but 20+ utility points less. Each control is one search; baselines get no search.

Our read · HarnessRSI

Score 68 · ★★ · evidence level: retained

Reports: held-out splits, transfer, ablations, cost ledgers.
Not reported: search variance, matched-budget baselines.